Counting Principles –
There are two basic counting principles, sum rule and product rule.

Sum Rule – If a task can be done in one of n_1 ways or one of n_2 ways, where none of the set of n_1 ways is the same as any of the set of n_2 ways, then there are n_1+n_2 ways to do the task.

Example: No. of ways to order a dish from 5 veg dishes and 6 non-veg dishes is (5+6) ways = 11 ways

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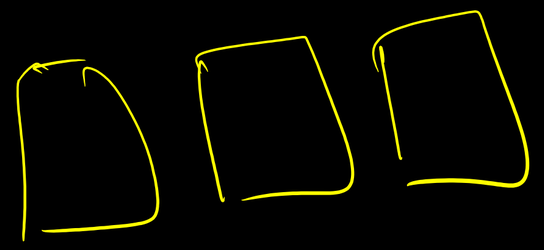
Example – In how many ways can a person choose a project from three lists of projects of sizes 10, 15, and 19 respectively?



$10 + 15 + 19 = 44$

Product Rule – If a task can be broken down into a sequence of k subtasks, where each subtask can be performed in $n_1, n_2, \ldots, n_k$ respectively, then the total number of ways the task can be performed is $n_1 . n_2 \ldots n_k$.

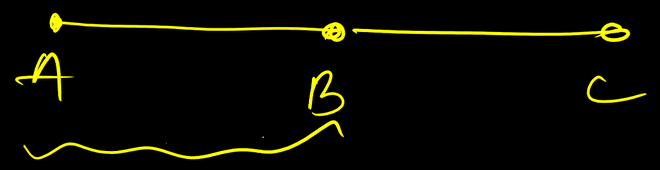
food	Drinks	Desert
15	10	5

$15 \times 10 \times 5 = 750$

Example – In how many ways can 3 winning prizes be given to the top 3 players in a game played by 12 players?

$$^{12}P_3 = \frac{12!}{9!} = 12 \times 11 \times 10$$

Problem 1 : There are three stations, A. B and C. Five routes for going from station A to station B and four routes for going from station B to station C. Find the number of different ways through which a person can go from A to C via B.



$5 \times 4 = 20$

Problem 2: A hall has 12 gates. In how many ways, can a man enter the hall through one gate and come out through a different gate.

12 Enter 11 Exit

$12 \times 11 = 132$